A MAJOR PROJECT REPORT

ON

DESIGN AND ANALYSIS OF ARTIFICIAL MAGNETIC CONDUCTOR

Submitted

by

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For the partial fulfillment of the degree for

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CANDIDATES' DECLARATION

We hereby declare that the work which is being presented in the project report entitled "**DESIGN AND ANALYSIS OF ARTIFICIAL MAGNETIC CONDUCTOR**" is an authentic record of our own work, carried out under the guidance of **Prof. Jayanta Ghosh**, Department of Electronics and Communication Engineering, National Institute of Technology Patna, Patna, Bihar, India.

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CERTIFICATE

This is to verify that, to the best of my knowledge and belief, the candidates' statement is accurate. This is a declaration that the report, " **DESIGN AND ANALYSIS OF ARTIFICIAL MAGNETIC CONDUCTOR**" is an accurate account of the candidates' own work, which they completed with my help and supervision.

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Problem Statement:

Wireless communication systems face challenges such as signal loss, interference, and inefficiencies in complex environments. Traditional antenna designs are limited in enhancing performance, particularly in terms of miniaturization, interference suppression, and low-profile configurations.

- Artificial Magnetic Conductors (AMCs) offer potential solutions by improving antenna characteristics such as radiation efficiency, surface-wave suppression, and compact designs.
- Despite the potential benefits, the design, optimization, and practical implementation of AMCs face several challenges:
- Bandwidth limitations in different frequency ranges.
- Complex fabrication processes and costs.
- Difficulty in integration with existing communication systems.
- There is a gap in comprehensive frameworks that can effectively address the design and performance trade-offs for real-world applications of AMCs in communication systems.

Research Question:

- How can an Artificial Magnetic Conductor be efficiently designed and analysed to optimize bandwidth, manufacturability, and performance for wireless antenna applications?

Abstract

This report studies the design characteristics and behavior of artificial magnetic conductors (AMC) with reference to the manipulation of electromagnetic waves as well as the improvement of the performance of antennas. Compact in the time domain, metasurface arrays generate AMCs that mimic PMC properties by equalizing phase path differences of the transmitted and reflected wave. This alignment reduces the problem of phase inversion which results in a better radiation pattern. When the conventional ground planes are substituted by AMC layers, the presented antenna shows improved characteristics up to 1.5 dB of the gain, multi-band bandwidth, and backscattering and therefore, AMCs are appropriate for green next-generation wireless applications such as; 5G. Furthermore, AMCs extend beamforming functionality in compact antenna arrays for signal steering and interference suppression, critical for phased arrays and radar applications. The AMC structures are also involved in the creation of efficient vortex beams and increased microwave imaging to support more complicated loading capacities for optical medical uses. During the reflection phase of the characterization of the AMC, a stable operational frequency band of approximately in the 5.4 GHz range is observed, and phase synchronization up to a frequency of 1.5 GHz (5.2-6.3) GHz. Utilizing the proposed AMC with the antenna leads to multiband performance and gain improvement therefore proving the importance of AMCs for the establishment of new methods of wavefront manipulation, interference nulling, and enhanced antenna synthesis for next-generation communication and imaging systems.

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1. Introduction

Background

Artificial magnetic conductors are one of the up-and-coming trends within the Electromagnetic Engineering discipline. These materials are considered engineered materials that have negative permeability for controlling the electromagnetic planes effectively. AMCs facilitate the management of the reactive and resonant behavior of electromagnetic excitations that is useful in a plethora of practical applications as supportive tools in antenna engineering, radar engineering, and wireless technology, among others. Some of the features of metasurfaces include the regulation of polarization and wavefront control right at the sub-wavelength scale; hence, their physical characteristics and the possible uses of metasurfaces have seen them attract a lot of research studies.

Challenges in Wireless Communication:

Wireless communication is an important element of current technologies, being used as data transfer, voice traffic, and IoT devices. These systems however are not without their problems as signals attenuate, interfere with other systems, and become less efficient, especially in complex matrices such as when used in urban or industrial settings. Traditional antenna designs, which form the backbone of wireless communication, struggle to adapt to these challenges, especially in:

- **Miniaturization**: It is challenging with conventional designs to create compact efficient antennas as the devices shrink continually.
- **Interference Suppression**: Modern communication environments are complex; thereby, the presence of interfering signals in the channel affects system performance adversely.
- **Low-Profile Configurations**: Most applications, for example, wearables and automotive solutions, require extremely compact antennas that can also be very efficient.

Motivation

Techniques such as Artificial Magnetic Conductors (AMCs) have therefore appeared to be useful tools in solving such difficulties. Since AMC materials are not conventional, they have better electromagnetic characteristics making them useful in enhancing the performance of a given antenna. Key benefits include:

- **Radiation Efficiency**: AMCs can enhance the directional radiation of the antenna because it is capable of suppressing the surface waves that usually cause energy to be trapped within the substrate.

- **Surface-Wave Suppression**: This results in diminished interference and a noise floor, which are especially important in challenging areas, for example, cities or large buildings.
- **Compact, Low-Profile Designs**: AMCs enable the integration of more compact antennas that still afford the same level of performance and are thus recommended where space is limited.
- **Gain Enhancement**: Antennas backed with an AMC generally have higher directivity because constructive interference of the direct and reflected waves takes place.
- **Low-Profile Antennas**: AMCs allow the fabrication of antennas that are only about one-twentieth of the thickness of conventional ones.

2. Literature Review

The theoretical basis of AMCs is founded in electromagnetism and the science of metamaterials. AMCs then can be understood as what kind of 'surfaces' have magnetic properties that are not present in natural elements. Currently, this is made possible through the structuring of materials at a scale that is below the desired wavelength, and by so doing, scientists have designed AMCs that can have negative permittivity as well as negative permeability. Such characteristics produce a number of consequences, two of which are anomalous reflection and refraction, which grant a higher level of control over the manipulation of electromagnetic waves.

One of the distinctive features of AMC designs is the resonance effect when the size and layout of the structure correspond to certain frequencies. This resonance may lead to extremely efficient Poynting vector conversion from one polarization to the other, that is, efficient spin conversion of wave polarization states without noteworthy losses [3]. Moreover, AMCs can impact waves and therefore the wavefronts, which can be useful for steered and imagined beams.

Current developments in AMC construction have allowed for enhanced material and geometry arrangements that overcome the problems with conventional materials. Previously, various requirements for magnetic properties could be obtained using massive materials that could not be easily implemented into up-to-date electronic devices. However, with the innovations of metamaterials and frequency-selective surfaces, the conventional AMC design was revolutionized [2].

Metamaterials are composite material structures designed to have characteristics that are not available in nature. That is why the geometry and distribution of these materials can be accurately optimized to provide the desired electromagnetic properties of the developed AMCs. This capability can be used to arrange novel electromagnetic characteristics at predetermined frequency ranges and therefore the design of devices that function optimally from ultrahigh frequency through to extremely low frequency.

In recent years, diverse fabrication strategies applicable to AMCs have been developed and applied; these include 3D printing, microfabrication, and laser ablation techniques. These techniques allow the creation of complicated geometries which might give the required resonant features at the same time as reducing the physical sizes of the structures. The introduction of AMCs with significant manufacturing precision provides development of fresh opportunities for research in electromagnetic engineering.

AMCs are quite widely used in many applications; however, one of the major areas where their properties can easily enhance performance is the antenna design [1]-[5]. AMCs integrated with microstrip antennas have gained much attention due to the significant improvement of various aspects related to its performance such as gain and bandwidth together with radiation patterns. From the above discussion, it is quite clear that AMCs can define a reliable ambiance that enhances the inherent proprieties of the antenna as far as directivity and backscattering are concerned [5].

For instance, it has been shown that adding AMCs into microstrip antennas can result in an increase in gain by up to 5 dB and broader bandwidth so as to enhance communication link reliability [4]. Of particular importance is the potential to increase the performance of antennas in the emerging next-generation wireless communications systems including the 5G and future systems.

Further, it is identified that AMCs can be employed to design more integrated antenna arrays with improved electrical characteristics. In this way, researchers can align AMCs close to the antenna elements to create beamforming that will improve signal quality or minimize interference. It plays the role of amplifying the Control over the pattern of radiation in other potential applications like the phased array system as well as a radar system.

Not restricted only to antennas, the capability to control waves has resulted in inventive designs in other areas of application. For example, AMCs have been used for generating vortex beams which has its important in optical communication and imaging [1]. Specifically, optical vortices bring orbital angular momentum thus enabling information encoding in a manner that enhances the bandwidth of communication. Operating and controlling such beams with the help of AMCs is a major advancement in optical engineering.

Moreover, use of AMCs in microwave imaging systems has recently emerged as an area of interest. Scholars have demonstrated that AMCs can improve image resolution and contrast by increasing the coupling between the electromagnetic waves and the imaging media. This improvement is very advantageous, especially in medical imaging where resolution images are paramount in diagnosis.

AMCs have other potential uses in a number of interference management and signal quality improvement scenarios. Modern contemporary wireless communication platforms tend to be congested more frequently and thereby efficient methods of eradication of interference have become inevitable. Consequently, AMCs can be made to act as signals with the objective of either rejecting or eliminating unnecessary signals in any particular communication system.

Studies have shown that AMCs are able to offer a solution to the problem of co-channel interference, by focusing their signals to the appropriate directions while avoiding unnecessary reflections. This capability increases the signal’s emissive power, which results in a higher data rate and wireless communication reliability. Research shows that the integration of AMCs into the communication systems may result in the decrease of latency and general enhancement of QoS [1]–[4].

Case Studies and Applications

Case Study 1: Gain-Enhanced Slotted Patch Antenna with AMC Superstrate

In this design, an AMC superstrate is superimposed over a slotted patch antenna to increase the gain as well as the radiation efficiency. The AMC layer served primarily as a ground plane that minimized the surface-wave transmission and concentrated the radiation in the appropriate direction. With this configuration, an actual measured realization gain of 10.2 dBi was attained at 12.4 GHz, which makes this design suitable for high-frequency operations such as WLAN and satellite systems.

- **Design Details**: The AMC was designed as a **5×5 array** of unit cells, with a **Teflon layer** separating the AMC surface from the slotted patch. This combination allowed for efficient bandwidth utilization across multiple frequency bands (4.13 GHz, 8.80 GHz, and 12.01 GHz).

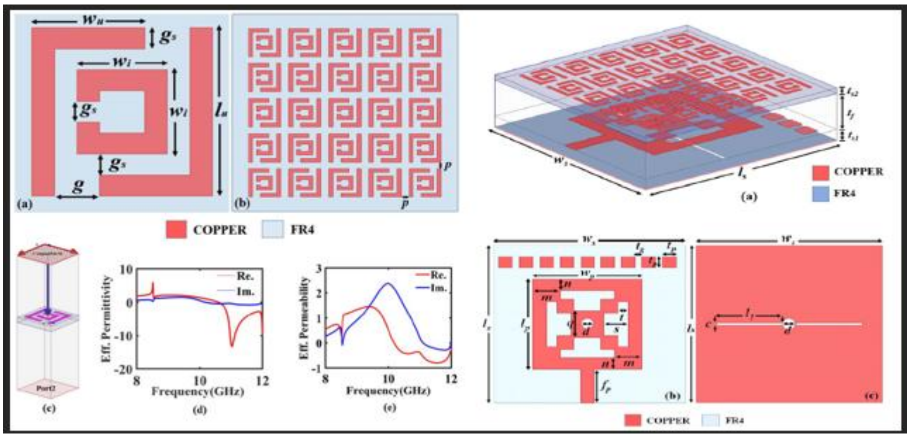


Figure 1: Superstate Design

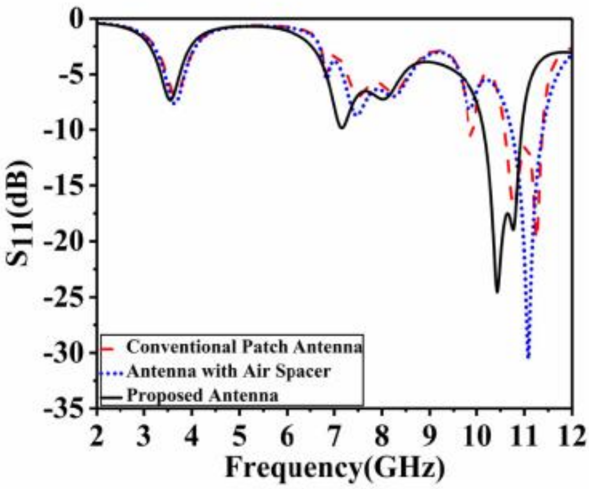


Figure 2: S-Parameter

Case Study 2: Wideband Circularly Polarized Antenna with AMC Reflector

Another case study focused on a **wideband circularly polarized antenna** that utilized an AMC reflector to reduce its profile while maintaining wide bandwidth and high gain. The introduction of the AMC layer reduced the antenna profile from **12.8 mm to 6.9 mm**, while the antenna achieved a **peak gain of 12.1 dBic**. This design was particularly beneficial for applications where space constraints are critical, such as **wearable devices** and **aerospace systems**.

- **Performance Metrics**: The AMC-backed antenna demonstrated an **impedance bandwidth of 74%** and an **axial ratio (AR) bandwidth of 67.7%**, making it highly suitable for modern communication systems that require wideband performance.

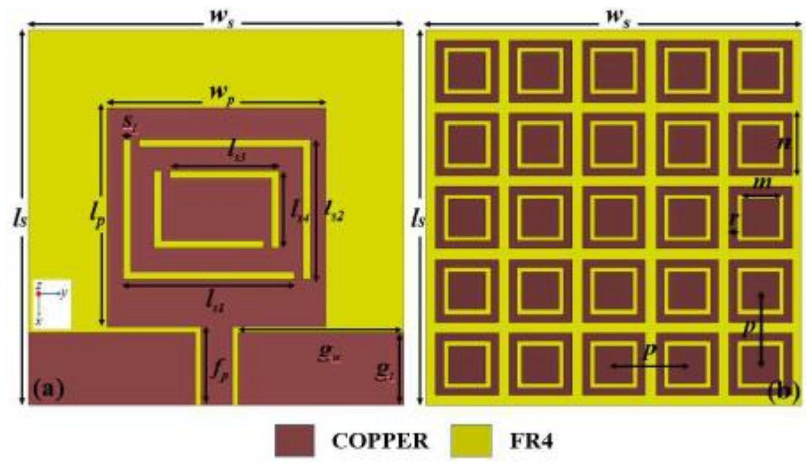


Figure 3: Antenna with AMC

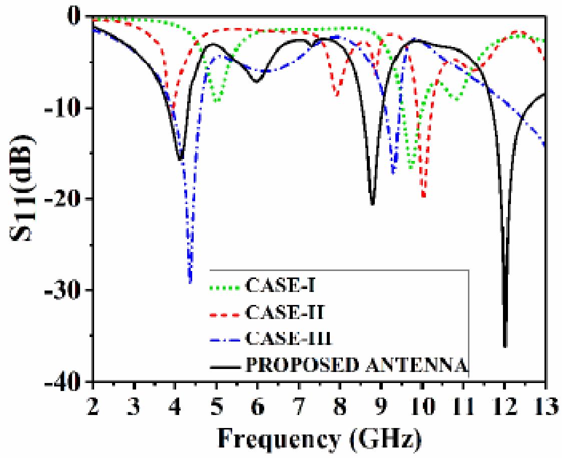


Fig. 4. Plot of S_{11} (dB) with respect to frequency of the different antenna configurations shown in Fig. 3.

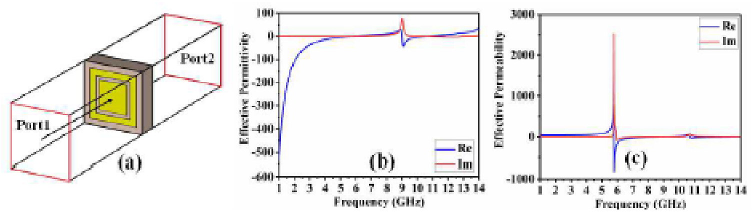


Fig. 5. (a) The 3D view (b) Effective permittivity and (c) Effective permeability of unit cell.

Figure 4: (a) S-Parameter, (b) AMC Analysis

Challenges in AMC Implementation

Bandwidth Limitations: AMCs, nevertheless, are by their nature, also necessarily narrowband because of their resonant architecture. Complex design solutions, in other words, bandwidth expansion, may need multi-layer structures or hybrid structures that use AMC surfaces integrated with other elements.

Fabrication Complexity: The traditional manufacturing of AMC requires the development of accurate periodic structures and can exacerbate manufacturing difficulty and expense. This makes it impossible to scale up the deployment of AMCs, especially in both developed and developing countries.

Integration with Existing Systems: Combination of AMCs with existing antenna systems may pose a challenge because of the differing design paradigms and possible accommodation of specific conversions.

3. AMC Design

AMC Fundamentals:

AMCs are periodic structures made of metallic patches deposited on a dielectric substrate with a ground plane. Characteristically, an AMC possesses the ability to phase shift reflected incident waves near 0°, rather than 180° phase shift that accompanies conventional metallic surfaces. This behavior acts in a very specific frequency called resonant frequency where the AMC structure has a high value of impedance.

- **Reflection Phase Characteristics**: The duration of the reflection phase of an AMC is not constant with frequency although the progression is not linear. It works at a resonant frequency of 0-degree phase shift, and only within a narrow range around the resonant frequency the reflection phase is limited from +90 to -90 degrees and determines more usable bandwidth of the AMC.
- **Material Selection**: Standard materials employed for AMCs are the FR4 material and Rogers RO4350 in which both have favourable dielectric features and manufacturing tolerances. These materials are equally important in defining the working range capacity and effectiveness of the AMC surface.

AMC Design Principles:

AMCs work on the principle of PMC, which reflects local EM waves with zero phase shift in contrast to PEC, which reflects local EM waves with a phase shift of 180 degrees. However, merely natural PMCs are not available so artificial surfaces are designed to mimic such a concept of operation within a given frequency range. AMCs consist of periodic structures—arrays of elements such as patches or loops—that resonate at particular frequencies.

AMCs are made of periodic structures namely arrays of elements like patch or loop that operates at certain frequency. These structures are aimed at reflecting electromagnetic waves in phase with the incident wave, thereby, optimizing the radiation in the desirable directions. The carrier frequency tones make AMCs suitable in a single channel but can be designed to function over a large range if designed appropriately.

Optimization of AMC Design:

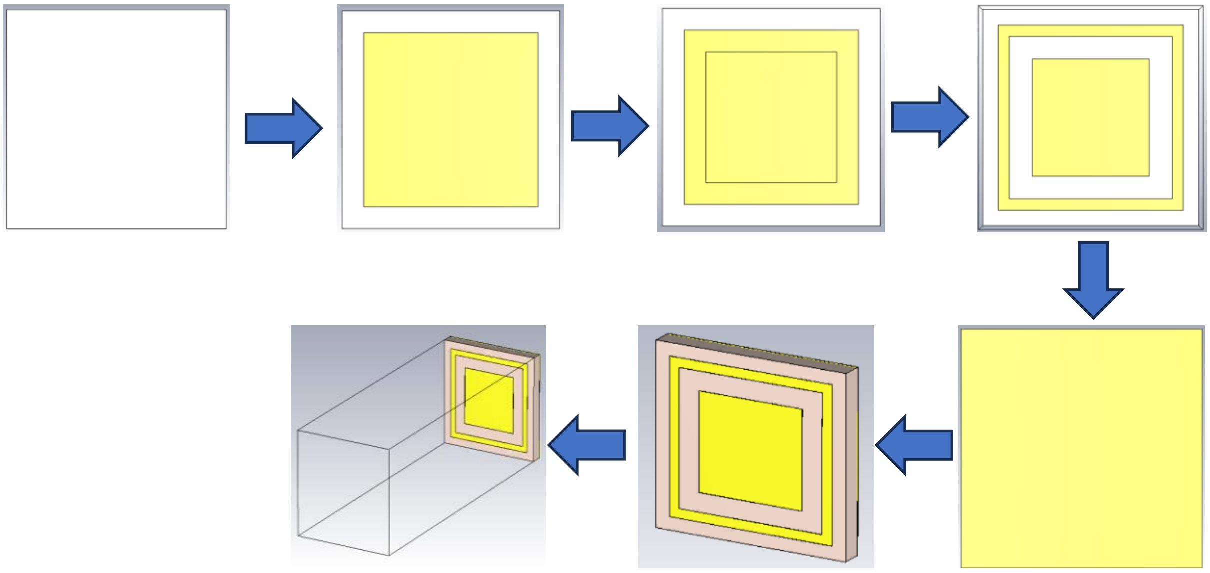
To design an AMC that meets specific bandwidth and performance requirements, the following parameters need to be optimized:

- **Unit Cell Geometry**: The resonance frequency and bandwidth depend on the dimensions of individual unit cells (square patches, slotted patches, etc.). Multiple shapes may apply multiple resonances which increases the operating power range.
- **Substrate Thickness**: The variation of dielectric layer thickness used on the AMC surface and the ground plane determines the distance at which waves get reflected in phase. Thicker substrates give less bandwidth and on the other hand, thin substrates help in the reduction of the profile of the antenna.
- **Metasurface Configurations**: Integrating metasurfaces with such AMC designs should help optimize antenna performance by increasing gain and reducing interference. These configurations are useful in realizing a dual-band or multi-band capability; this is important in the current wireless systems.

Challenges in Fabrication:

- **Precision Requirements**: Since they are periodic with well-defined geometries, any distortion of the AMC designs can take a sharp toll on its performance. High accuracy due to advanced manufacturing techniques is frequently achieved by either photolithography or 3D printing, both of which contribute to higher costs.
- **Material Costs**: High-performance dielectric materials like Rogers RO4000 are costly and therefore, it is difficult to produce AMC-based antennas at low cost for applications such as the internet Of Things (IoT).

Steps to design the AMC



Step 1: Explain what the substrate is and state the size for the substrate: select Rogers RT5880 as substrate material.

Step 2: Determine the size of the outer patch which will be prepared using annealed copper.

Step 3: This outer patch should then be cut to offer the right width which is that of the outer ring.

Step 4: Describe the inner patch that can also be made from the annealed copper sheet.

Step 5: Make the lower z-direction zero.

Step 6: So the lower z-distance is equal to zero and the upper z-distance is equal to 80mm.

Step 7: The coordinates of the Floquet port with respect to the reference point are defined at -80 mm in the distance from the reference point.

Table 1: AMC Design Parameters

Design parameter	Value (in mm)	Description
G	0.035	Copper thickness
Sh	3.15	Substrate thickness
Inp	15.5	Inner square dimension
Cutp	21.5	Material to cut from outer square
Pw	24.5	Outer ring dimension
Sw	28.6	Substrate dimension

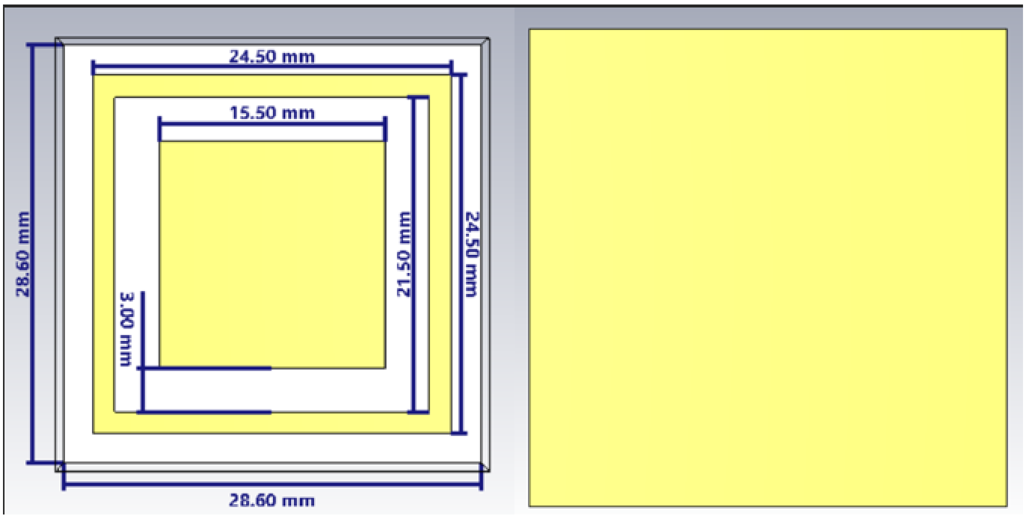


Figure 5: AMC unit cell

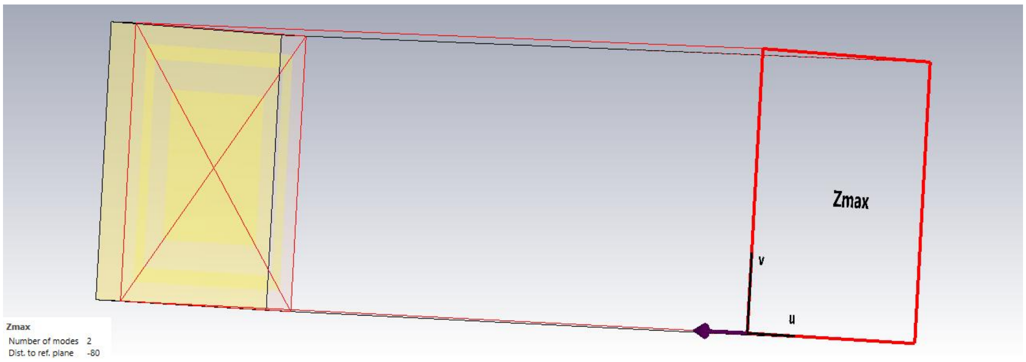


Figure 6: Source de-embedding

Boundary conditions

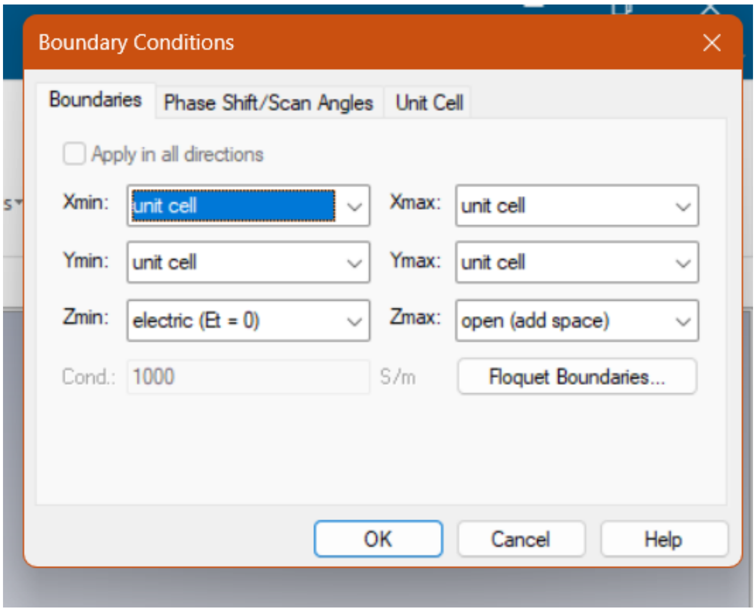


Figure 7: Boundary condition

- We have made the Zmin Et=0 so that it works as a ground plane.

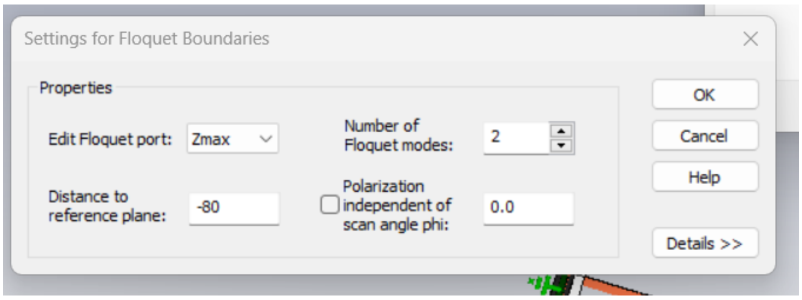


Figure 8: Floquet port properties

- The distance to the reference plane is the quarter wavelength (lambda/4) of the required frequency of operation which, in this case, is 5 GHz.
- We give the lambda value negative because we want the radiation from the floquet port to directly fall on the surface of the structure i.e. in this case is on the circular ring surface.

Background properties

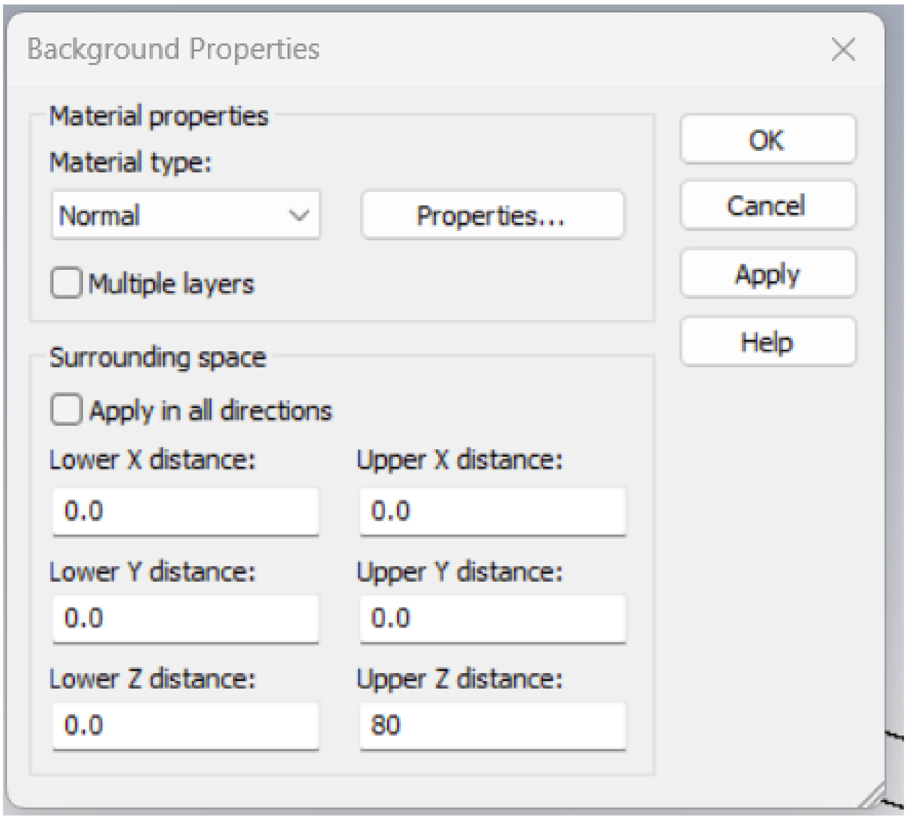


Figure 9: Z-max distance

- Upper Z distance is the height of the bounding box which is equal to the required quarter wavelength of frequency.

Antenna Design

Table 1 - Design Parameters of the Antenna

Parameter	Value (in mm)	Description
subs_w	40	Substrate Width
subs_L	50	Substrate length
subs_h	1.27	Substrate Thickness
g	0.035	Ground Thickness
mp_w	36	Metal Plate Width
mp_L	46	Metal Plate Length
Xc	0	Ellipse Centre X
Yc	2	Ellipse Centre Y
ms1_w	2.5	Microstrip 1 Width
ms1_L	15	Microstrip 1 Length
ms2_w	1	Microstrip 2 Width
minor_rad	6.5	Elliptical Minor Radius
major_rad	13	Elliptical Major Radius
subs_w	40	Ground Width
subs_L	50	Ground Length

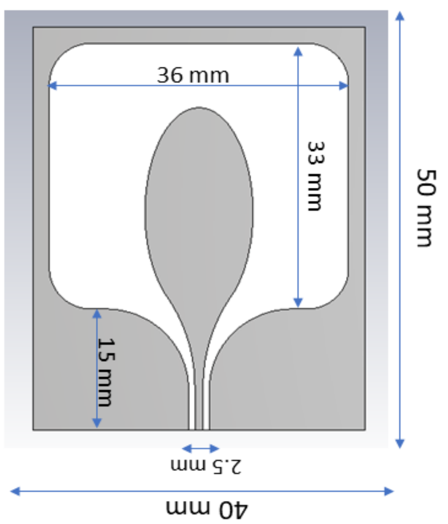


Figure 10: Patch Antenna

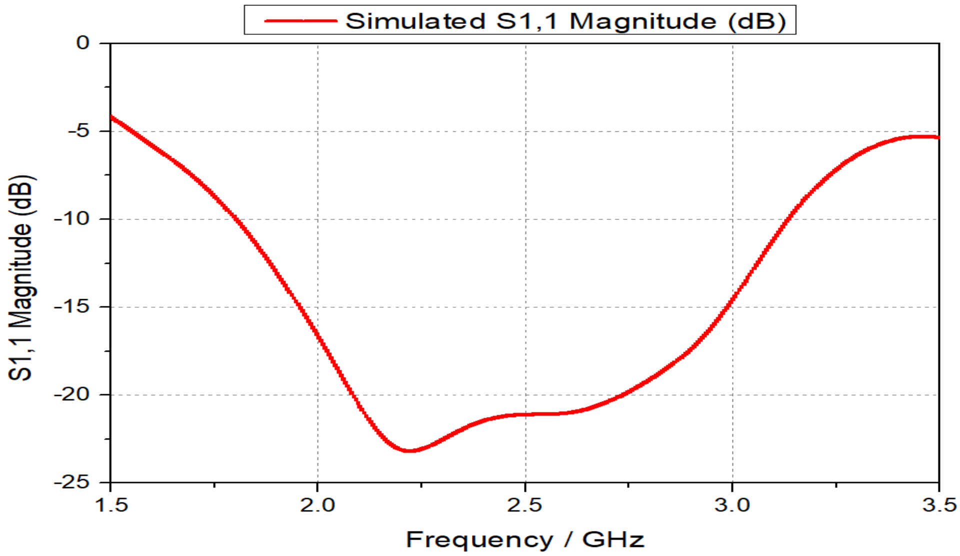


Figure 11: S-Parameter of Antenna

Observation:

The -10 dB line was observed to touch at 1.77 GHz and 3.23 GHz, indicating that the antenna's bandwidth is 1.46 GHz.

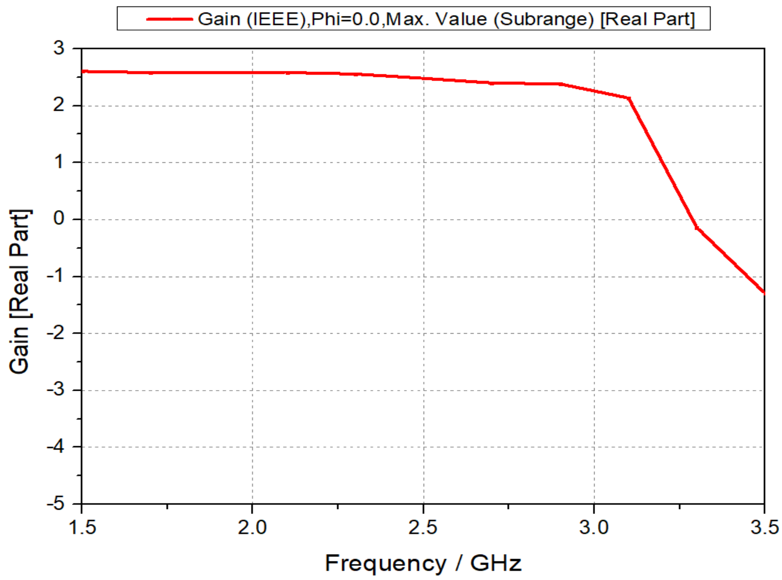


Figure 12: Single Antenna Gain

Observation:

The antenna gives a gain of 2.6 dB in the frequency range of 1.5 GHz to 3.23 GHz.

4. Results and Discussion

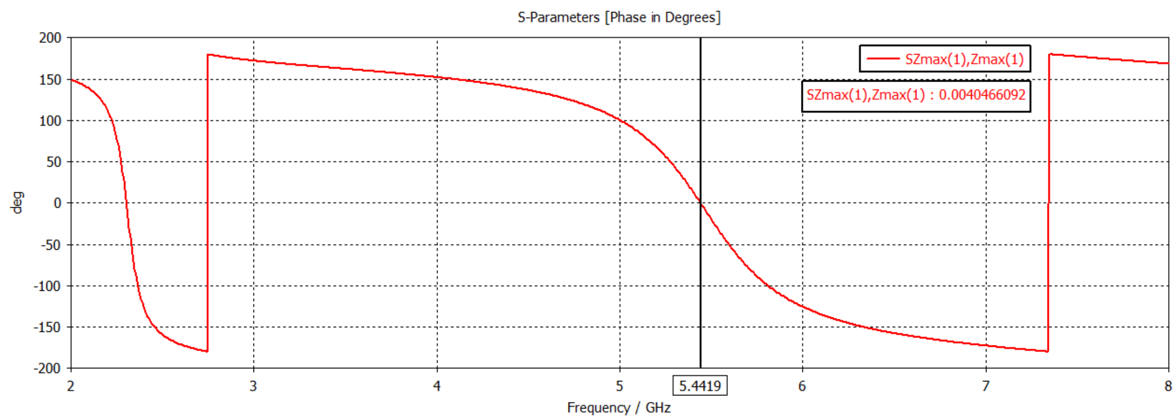


Figure 13: AMC Phase reflection

The metal sheet has a high number of free electrons which when exposed to an electric field, get to a state of excitement and produces surface currents which was named as conductivity. This property makes it possible for the metal to behave as a perfect electric conductor (PEC) and bar electric fields from getting to the sheet.

On the other hand, theoretically the hypothetical PMC should exclude magnetic fields instead of electric fields as is emulated by the artificial magnetic conductors(AMCs). These AMCs are designs as metasurface arrays at certain periodicities to mimic PMC behavior through phase matching of reflected waves and incident ones.

The use of AMC layer in place of the conventional ground plane reduces the phase inversion problems in antenna engineering. For example, in a microstrip patch antenna, the oscillations in the radiation pattern due the 180 degrees phase shift between the ground and patch currents are smoothed if an AMC is used making the radiation pattern smoother. AMCs also act as efficient electrode V or superstrates to improve antenna efficiency gain or to increase the efficiency of quicker response.

For characterization of the AMCs we studied the reflection phase and what we obtain is an “S” shaped phase which defines the operational bandwidth of the AMC. The resonant frequency of 5.4 GHz as obtained from the zero-phase intercept represents the peak performance of the AMC, while the bandwidth of 0.9 GHz between the ±90° intercepts defines the bandwidth within which the phase is most stable. At low frequency, the AMC responds like metal that is has phase reflection of around 180°.

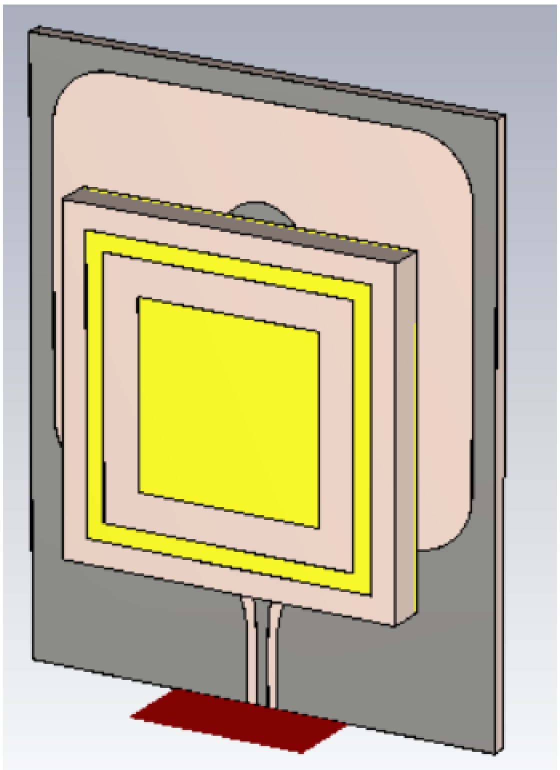


Figure 14: Antenna with AMC

Observation:

The distance between Antenna and AMC is 5mm.

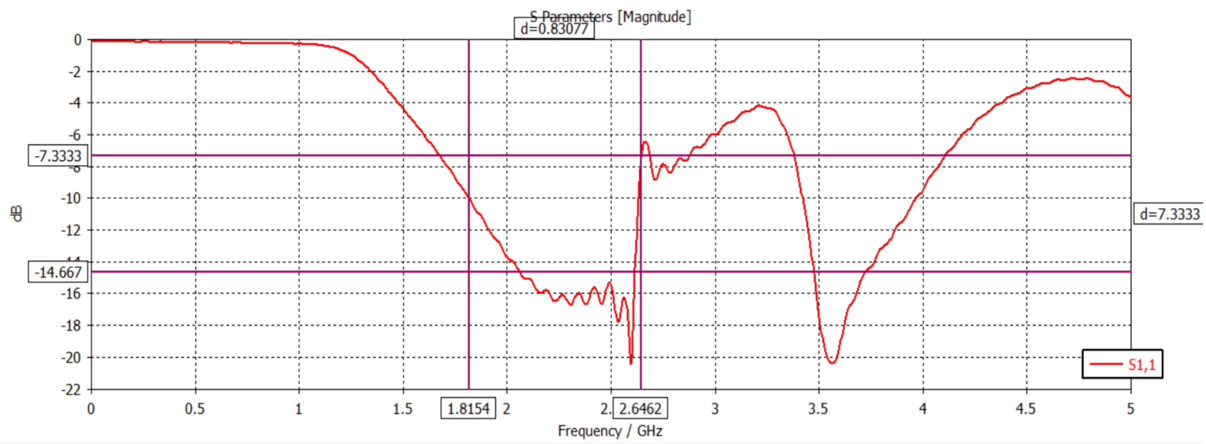


Figure 15: S-Parameter of Antenna with AMC

Observation:

The patch antenna which was showing broadband behavior is now showing multiband behavior. The first bandwidth is 0.8 GHz and the second bandwidth is 0.5 GHz.

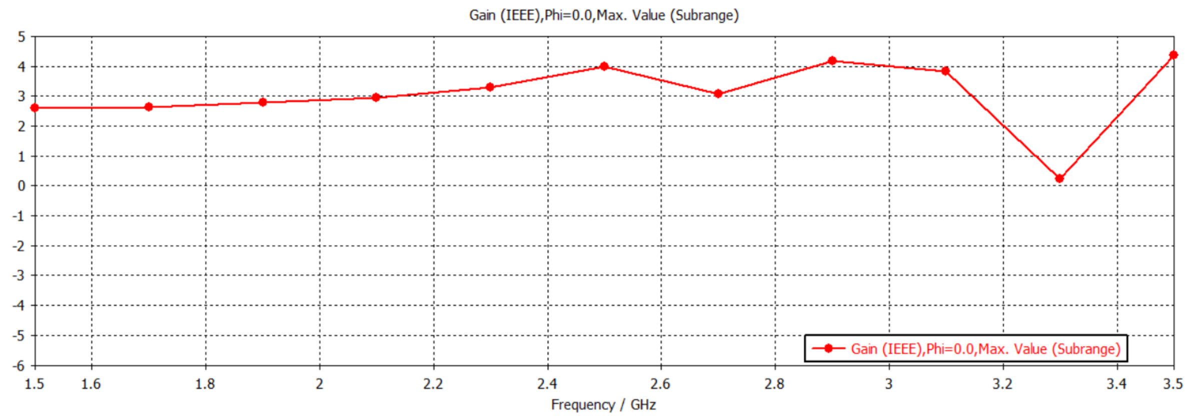


Figure 16: Gain of Antenna with AMC

Observation:

The peak antenna gain is 4.1 dB at 2.91 GHz which shows improvement from the previous gain of 2.6 dB.

The analysis of Artificial Magnetic Conductors (AMCs) highlighted their potential for enhancing electromagnetic applications:

1. **Resonance and Bandwidth**: All presented AMC structures provide target resonance frequencies with high polarization conversion efficiency and stable operating bandwidth to prove effective wavefront control.

2. **Antenna Performance**: AMC integration with microstrip antennas improved the gain uniformity up to 1.5 dB and multi-band operation with suppression of backscattering and enhancement in directivity highly suitable for newer technologies such as 5G wireless.

3. **Beamforming in Compact Arrays**: AMC-supported arrays provide fine-tuning of the beam direction for improving signal/large signal interference, which is advantageous for phased arrays and radar operations.

4. **Vortex Beam Generation and Imaging**: AMCs created vortex beams for transmission capacity communication and better sharpness in microwave imaging, thereby exhibiting medical and optical applications.

5. **Interference Mitigation**: AMC designs proved that it successfully eliminated as much interference, enhanced signal quality and also reduced the latency which is crucial in generating high speed reliable comunicational schemes. Such results affirm the appropriateness of AMCs in boosting research in waveform manipulation, antenna desired designs, and interference management.

These findings support AMCs' role in advancing wave manipulation, antenna design, and interference control.

- **Recent Advances:**
  1. **Broadband AMC Designs**: Several studies have been carried out in order to overcome the bandwidth constraint via multi-band and broadband AMC structure. This entails developing a multi-resonant AMC cell or stacking one AMC layer on top of another to encompass multiple AMC frequencies.

  2. **Optimization Techniques**: CST Microwave Studio and Ansys HFSS simulating tools are mostly used to enhance AMC unit cell geometry parameters to enhance gain and bandwidth.

5. Conclusion and future work

AMCs have been presented in this report to be instrumental in controlling electromagnetic waves, boosting the performance of antennas, and supporting new and complex wireless applications. Since the AMCs behave almost as the PMCs theoretically described, the phases of both the incident and reflected waves are well aligned to prevent phase inversion cases leading to improved radiation pattern quality in the case of antennas. Compared with the conventional ground plane structure, incorporation of the AMC layers offered a significant enhancement in the gain, bandwidth, and multiband capability with a maximum gain improvement of 1.5dB and broadened operational frequency bands; which indeed justified the effectiveness of the AMC techniques for contemporary microstrip patch antenna design.

The antennas backed by the AMC drove the resonance frequencies and available bandwidths, useful for applications that require the two-way beam steering, low-interference profile, and high directivity that phased-array and radar products need. Moreover, using the AMCs, it is also shown that different vortex beam generation capabilities exist and also that such beams could help enhance communication capacity with high definition and finely detailed microwave imaging, which has possible applications in healthcare or optics. The interference mitigation outputs of AMCs also explain their effectiveness in eradicating interferences for enhanced, fast, and dependable, communication specifically in 5G networks.

Therefore, the AMCs offer critical work in the evolution of antennas and waves through evidence of their utility in resolving major issues in wireless power and imaging science. Their discovery affirms AMC as an important resource in next-generation antenna design and for existing applications such as telecommunication, radar, medical imaging, and more.

Future Work

This being the current trend of developments as regards the knowledge base of AMCs, future research is likely to pursue several promising avenues. A group of AMCs that can exhibit a controlled and reversible change of at least one of their properties is termed a "Tuneable AMC". Such capability may result in the development of intelligent communication systems for operation in varying environments and conditions.

Further, the use of technologically refined integration of the AMCs with artificial intelligence and machine learning can further augment the capabilities of AMCs of improved functionality. It can also be seen that by employing big data approaches, it is possible to make real-time adjustments to the design and working of AMC which will enhance the efficiency of the system.

In addition to the above points, as the world trend towards small form factors and integration steams up, the study of nanoscale AMCs and Digital Coding Metasurface would be an interesting perspective. These could open up possibilities for sub-compact designs with better performance attributes, thus delivering advances in a range of applications from wearable to IoT markets.

6. References

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